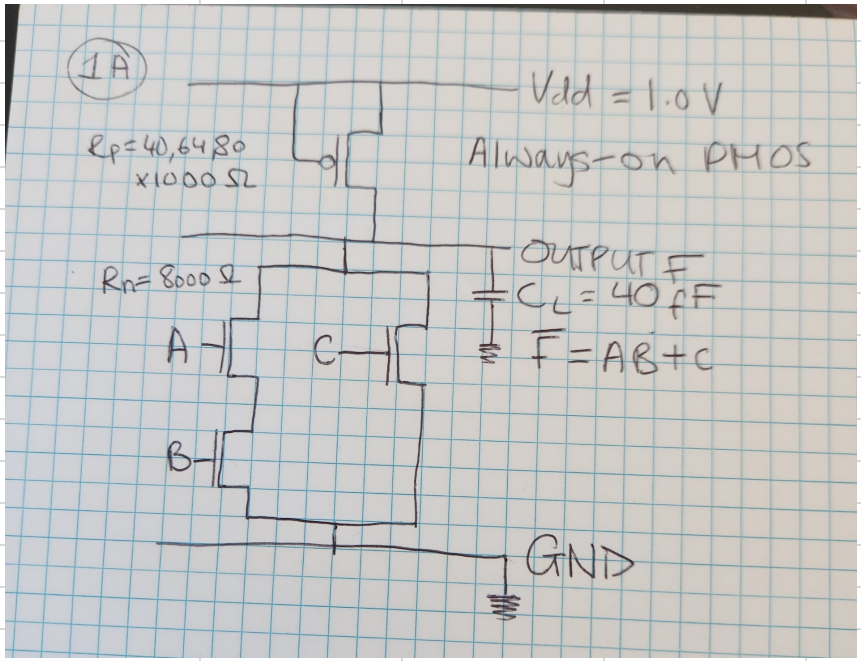
1A



I apologise for the trouble! Due to my disability, I want to preserve my wrist and hand function so I am typing the work I can. I hope that I have formatted my work in a clear enough manner to mark, but if not, please let me know and I will explain my reasonings over a Zoom call or in person.

VOL ≤0.20 V

Vol = Vdd*Rn/Rtotal

PMOS Rp (ohms)	Vdd (V)	CL (fF)	ABC	Notes	Ohms	Vol
40000	1	40	110	AB sequence, 2Rn.	16000	0.2857142857
40000	1	40	`011`	C sequence, Rn	8000	0.1666666667
40000	1	40	111	AB and C,	5333.333333	0.1176470588

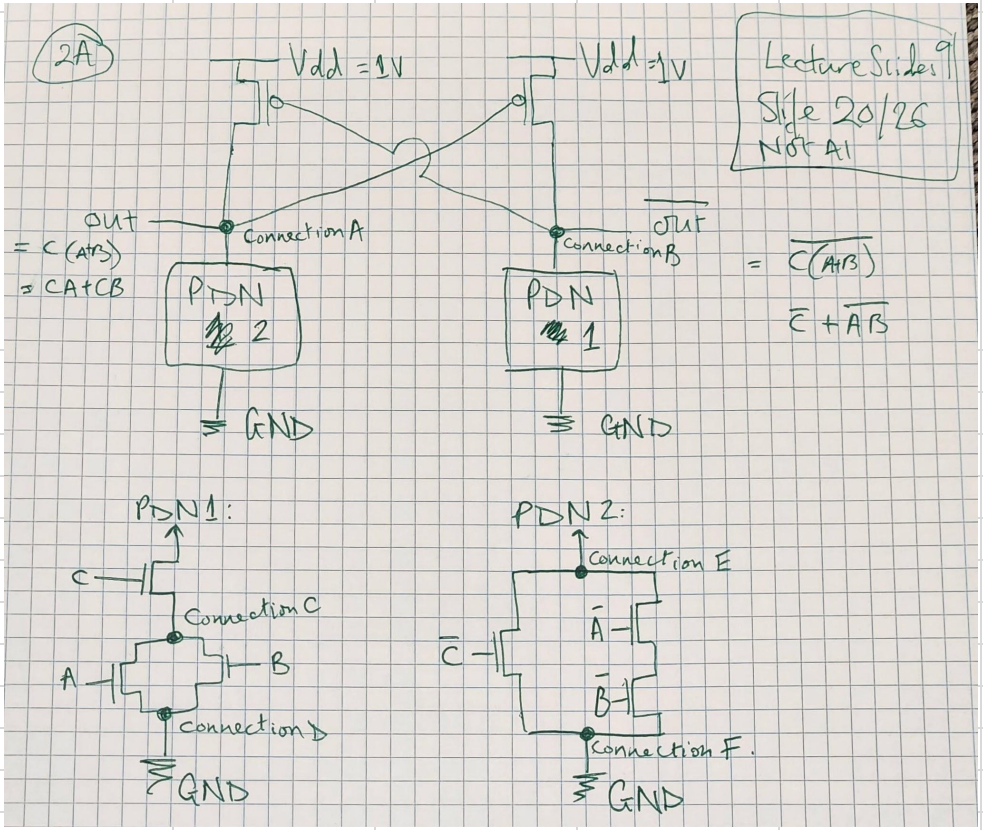
PMOS Rp (ohms)	Vdd (V)	CL (fF)	ABC	Notes	Ohms	Vol
64000	1	40	110	AB sequence, 2Rn.	16000	0.2
64000	1	40	`011`	C sequence, Rn	8000	0.1111111111

	64000	1	40	111	AB and C,	5333.333333	0.07692307692
	PMOS Rp (ohms)	**Vdd (V)**	**CL (fF)**	**ABC**	**Notes**	**Ohms**	**Vol**
	80000	1	40	110	AB sequence, 2Rn.	16000	0.1666666667
	80000	1	40	`011`	C sequence, Rn	8000	0.09090909091
	80000	1	40	111	AB and C,	5333.333333	0.0625
	The pull up resistance of 80000 perfectly satisfies the Vol requirement of 0-outputs having a Vol output of less than 0.2. As the requirement is that Vol is less than or equal to 0.2, 64000 ohms also squeaks by.						
	Pstatic = Vdd*I		V=IR	I=V/R	P = 1*1/R		
	PMOS Rp (ohms)	**Ohms**	**Vol**	Vdd	P (watts)		
	64000	5333.333333	0.07692307692	1	0.000014423076		14.4 Microwatts
	PMOS Rp (ohms)	**Ohms**	**Vol**	Vdd	P		
	80000	5333.333333	0.0625	1	0.00001171875		11.7 Microwatts
	Vol is affected by the ratio between resistors - think of the VTC curve! - and static power is over the whole circuit and depends on all the resistors summed.						
1B							
	A=1	C=0	B = 0 --> 1 --> 0		Circuit transition from on to off to on		
	Pull up resistances: 64000 and 80000 Ohms				Vdd --> Vol --> Vdd because PMOS always on		
					100-->110-->100		
	FALLING PROPAGATION: ON TO OFF						
	Resistance pulldown = 16000 ohms, 8000 + 8000				Vout(t) = Vpost+ (Vpre-Vpost)*e^(-time/tau)		
	Rp active, Rn active.	Equivalent resistance:	(RP*RN)/ (RP+RN)				

PMOS Rp (ohms)	**Ohms**	R equiv	Vol	Vdd	CL	Tau
64000	5333.333333	4923.076923	0.2	1	40*10^-15	0.0000000001969230769
		Ohms				
0.5 target V	equals	0.2 + (1-0.2)*e^(-time/tau)				
0.3/0.8	equals	e^(-time/tau)				
-0.980829253	equals	-time/tau				
0.0000000001931479144		=time	equals	193 picoseconds		
PMOS Rp (ohms)	**Ohms**	R equiv	Vol	Vdd	CL	Tau
80000	5333.333333	5000	0.1666666667	1	40*10^-15	0.0000000002
		Ohms	V			
0.5 target V	equals	1/6+ (1-1/6)*e^(-time/tau)				
2/5	equals	e^(-time/tau)				
-0.9162907319	equals	-time/tau				
0.0000000001832581464		=time	equals	183 picoseconds		(unsure)
RISING PROPAGATION: OFF TO ON						
Current depends only on always-on pmos and the nmoses turning off						
PMOS Rp (ohms)	**Ohms**	R equiv	Vol	Vdd	CL	Tau = Rp CL
64000	5333.333333	4923.076923	0.2	1	40*10^-15	0.00000000256
Vout	equals	Vfinal + (Vpre - Vpost) * e^(-time/tau)				
0.5	equals	1+(0.2-1)*e^(-time/tau)				
-0.5/(-0.8)	equals	e^(-time/tau)				
0.4700036292	equals	time/tau				

0.000000001203209291			equals	1203 picoseconds		
PMOS Rp (ohms)	**Ohms**	R equiv	Vol	Vdd	CL	Tau = Rp CL
80000	5333.333333	5000	0.1666666667	1	40*10^-15	0.0000000032
Vout	equals	Vfinal + (Vpre - Vpost) * e^(-time/tau)				
0.5	equals	1+(1/6-1)*e^(-time/tau)				
-0.5/(5/6)	equals	e^(-time/tau)				
0.5108256238	equals	time/tau				
0.000000001634641996		equals	1634 picoseconds			
Delay:						
For 64k ohms, falling delay is 193 picoseconds and the rising delay is 1203 picoseconds. For 80k ohms PMOS, falling delay is 183 picoseconds and rising delay is 1634 picoseconds. For a timing-critical application, I would go for the smaller PMOS resistor to minimise the rising time as the difference between PMOS resistors there is far greater than the difference in falling times.						
Static Power:						
For the worst case, where the path of current is two resistors in series, the 64k ohm PMOS loses 14.4 microwatts and the 80k ohm PMOS loses 11.7 Microwatts. For a timing constrained circuit, I would choose the bigger PMOS transistor as the amount of power lost is less.						

2A

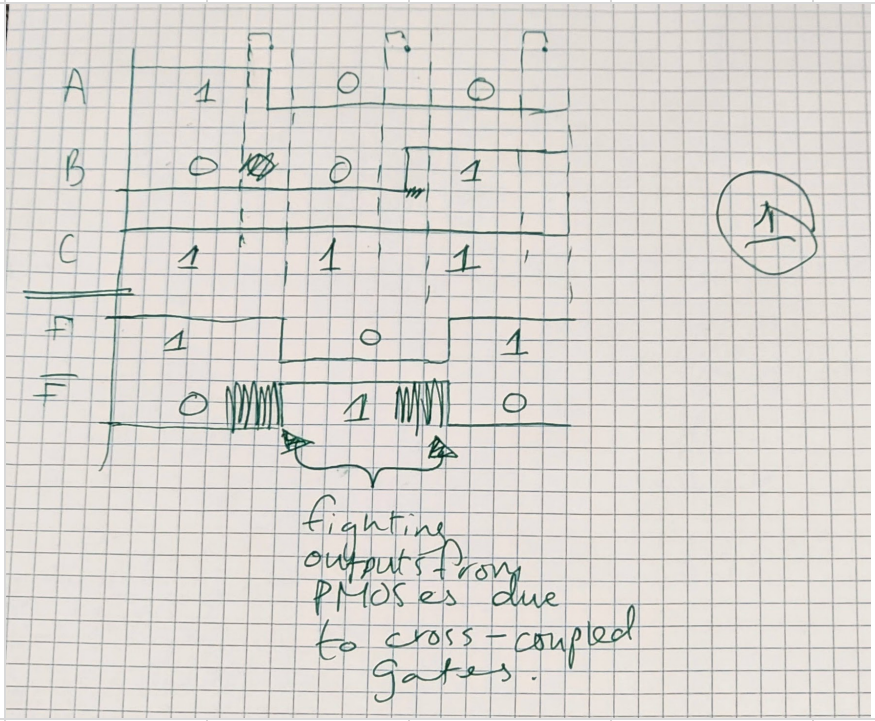


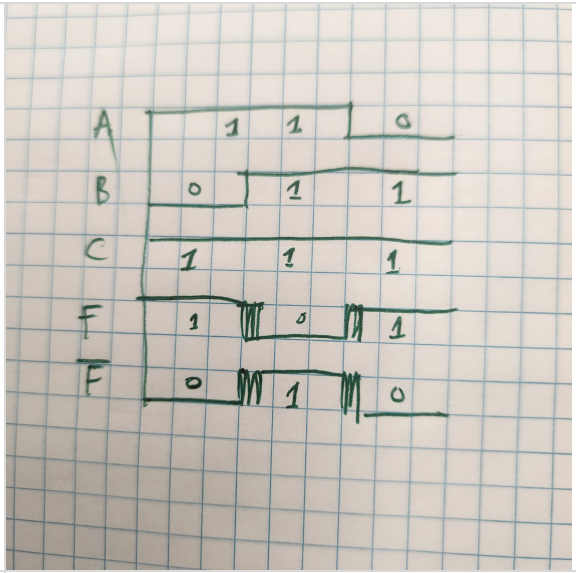
AI used for comprehension of topic here, not answer of problem (as suggested by Prof Joshi). I am not confident this answer is correct at all.

F=CA+CB

ABC = 101	ABC = 001
PDN1 conducts, PDN2 does not.	PDN2 conducts, PDN1 does not.
PDN1 links to ground, PDN2 does not.	PDN2 connected to ground makes F 0.
F = 1, Fbar = 0	Fbar = 1

Both outputs can't stably rest at 1, because the two PDNs are strictly complementary. If the output is one, then that means that the PDN associated equals zero, which means that the other PDN would pass current through, which would drain the voltage from the alternate current. The two PMOSes make a vicious negative cycle.

2B	Scenario I: A : 1 →0 occurs before B : 0 →1.						
	(A,B,C) = (1,0,1) to (0,1,1), with C = 1 throughout.						
	States:	101	"001"	"011"			
							
	Scenario 2: B: 0 →1 occurs before A : 1 →0.						
	(A,B,C) = (1,0,1) to (0,1,1), with C = 1 throughout.						
	States:	101	111	"011"			



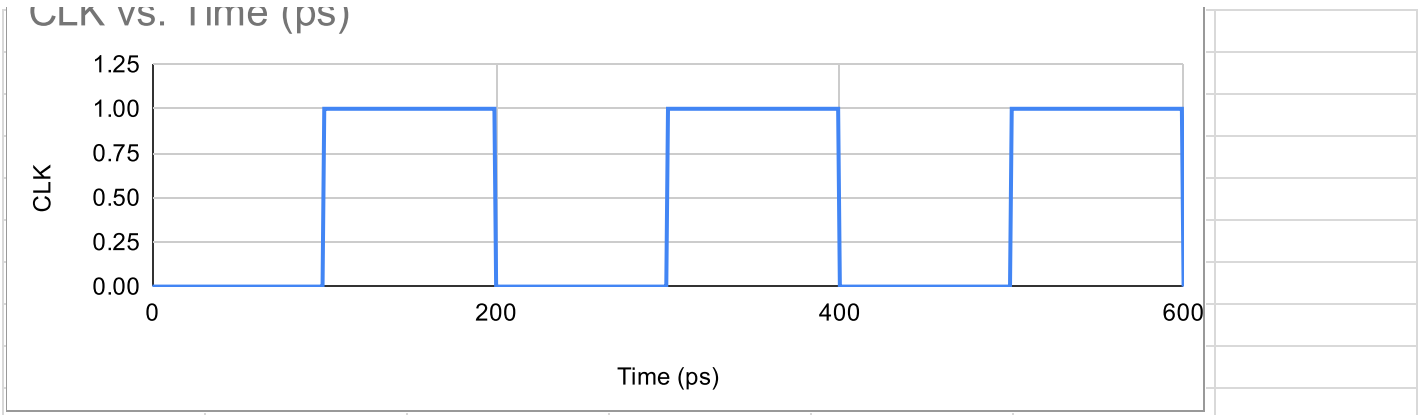
The uncertainties here also stem from the fighting cross-coupled PMOSes.

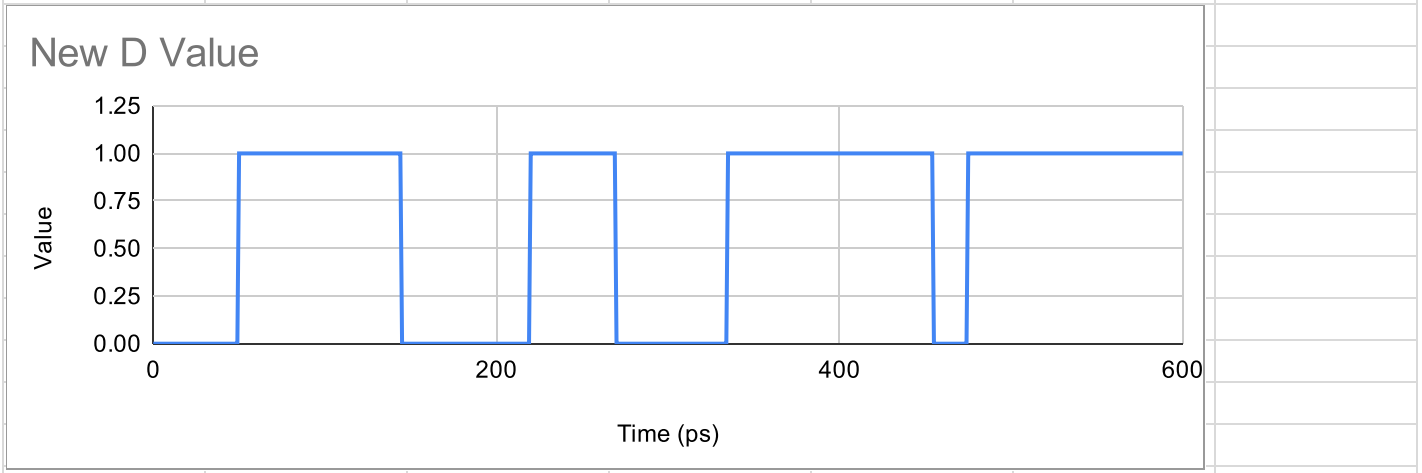
To solve this issue, the circuit could introduce a clock? I'm not sure how sensible an idea this is though. It would lock values to pause propagation which would cause values to be sampled not pingponged.

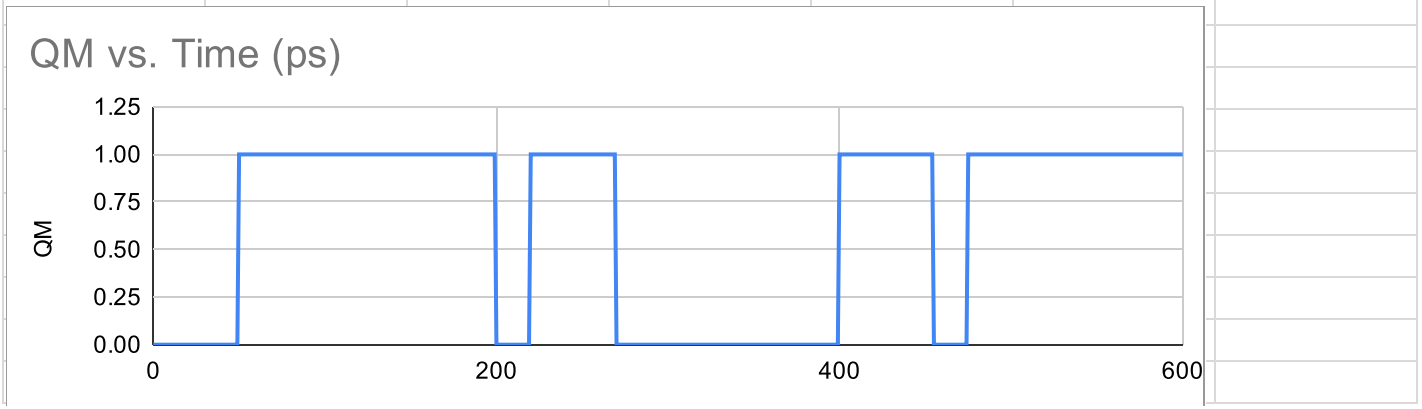
3A

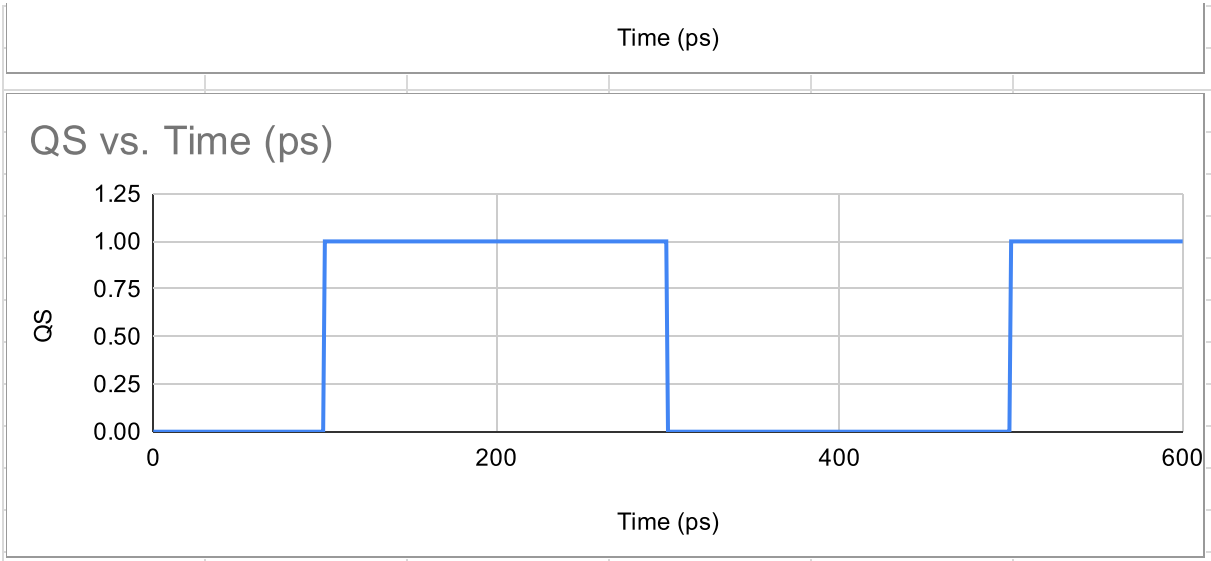
Time (ps)	CLK	New D Value	QM	QS
0	0	0	0	0
49	0	**0**	0	0
50	0	**1**	1	0
99	**0**	1	1	0
100	**1**	1	1	1
144	1	**1**	1	1
145	1	**0**	1	1
199	**1**	0	1	1
200	**0**	0	0	1
219	0	**0**	0	1
220	0	**1**	1	1
269	0	**1**	1	1
270	0	**0**	0	1
299	**0**	0	0	1
300	**1**	0	0	0
334	1	**0**	0	0
335	1	**1**	0	0
399	**1**	1	0	0
400	**0**	1	1	0
454	0	**1**	1	0
455	0	**0**	0	0
474	0	**0**	0	0
475	0	**1**	1	0
499	**0**	1	1	0
500	**1**	1	1	1
599	**1**	1	1	1
600	**0**	1	1	1




CLK vs. Time (ps)
1.25
1.00
0.75
0.50
0.25
0.00
CLK
0
200
400
600
Time (ps)

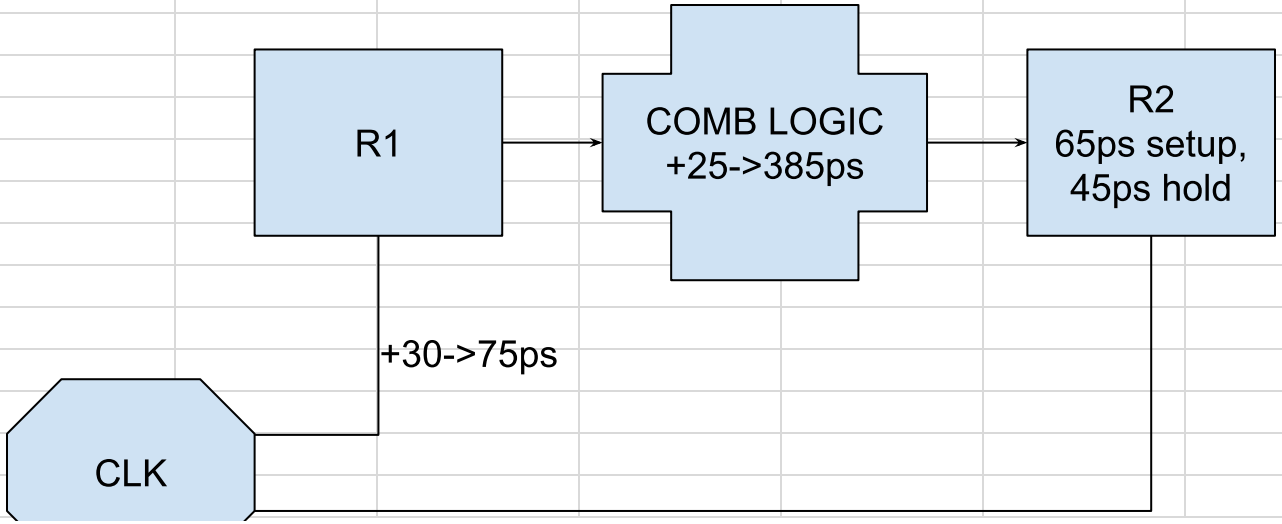

New D Value
1.25
1.00
0.75
0.50
0.25
0.00
Value
0
200
400
600
Time (ps)


QM vs. Time (ps)
1.25
1.00
0.75
0.50
0.25
0.00
QM
0
200
400
600



A change in D when the clock is high is not caught by the 'master', and so therefore is not observed by the 'slave'. The 'slave' observes the 'master' latch only, I believe?

3B

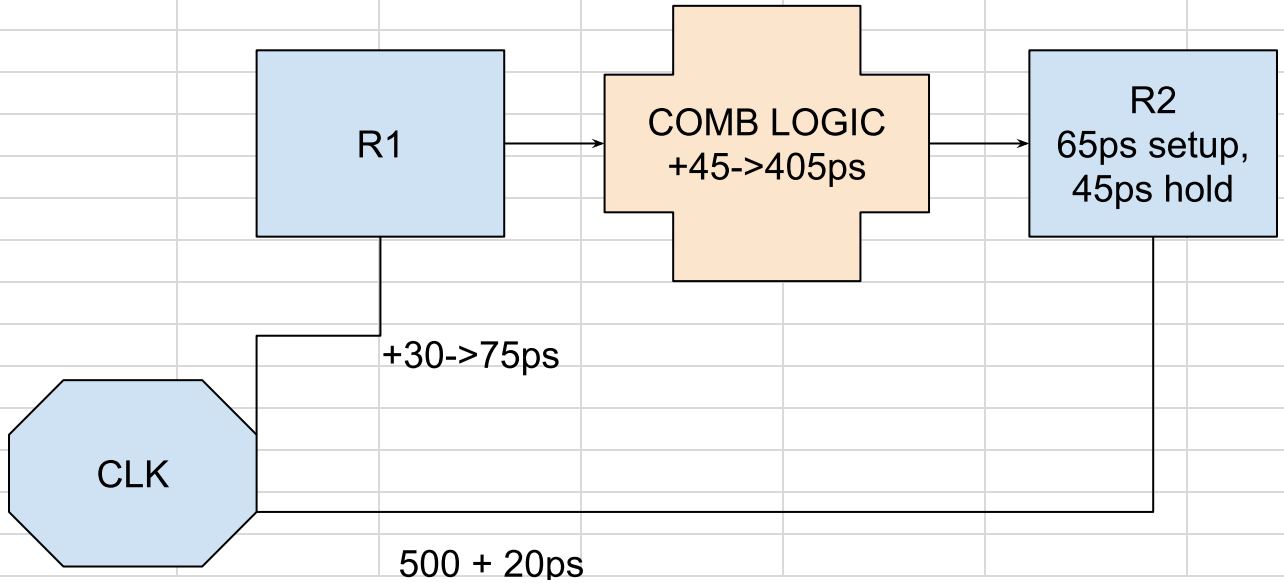




500 + 20ps

Maximum delay of data getting to R2:			It is possible that data would be output 5ps after R2 is ready.
75+385	equals	460	
Minimum time needed for R2 to be ready to accept data:			
500+20-65	equals	455	
Minimum delay of data getting to R2:			
30+25	equals	55	Data could arrive too soon and overwrite R2's inputs
Maximum time R2 needs data from previous clock:			
20+45	equals	65	

Added delay buffer to COMB LOGIC



Maximum delay of data getting to R2:				It is possible that data would be output 25ps after R2 is ready.		
75+405	equals	480				
Minimum time needed for R2 to be ready to accept data:						
500+20-65	equals	455				
Minimum delay of data getting to R2:				Data will arrive with a 10ps buffer. Hold time error fixed		
30+45	equals	75				
Maximum time R2 needs data from previous clock:						
20+45	equals	65				
Hold time errors are not affected by clock time? The error here was fixed, though worsening the set-up error, by increasing the time of the combinational logic block?						

4A							
	Didn't attempt - no time after dealing with DASS/Testing center.						
4B							
	Didn't attempt - no time after dealing with DASS/Testing center.						
AI STATEMENT							
	Did not use, except to explain concepts (as advised to try by Prof Joshi)						